

Neural Networks and Deep Learning

This concise module is designed for AI-agent based learning and quick revision.

Artificial Neural Networks

Inspired by the human brain and composed of interconnected neurons.

Perceptron

The basic building block of neural networks.

Activation Functions

ReLU, Sigmoid and Tanh introduce non-linearity.

Feedforward Network

Information flows from input to output.

Backpropagation

Updates weights using error gradients.

Gradient Descent

Optimization algorithm to minimize loss.

CNN

Specialized for image processing.

RNN

Processes sequential data such as text and speech.

Quick Revision: Revise definitions first, understand key concepts, then practice MCQs.